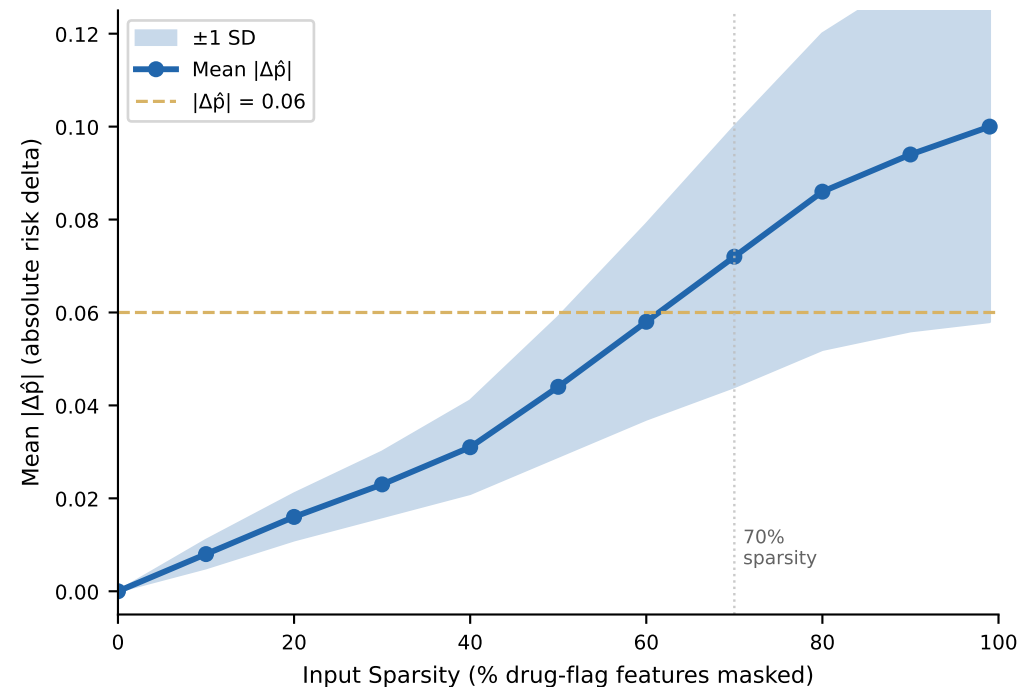


Imputation of Normality — Sensitivity Analysis

Young-Adult Opioid Cohort, Low Density Bin (n=2,000 sampled)

(A) Prediction Error vs. Input Sparsity
(1,000 bootstrap trials; 2019 holdout)



(B) Risk-Band Stability vs. Sparsity
(Low/Moderate/High band preserved vs. full input)

